

**Multiple Choice:**

1. In a parallel ac circuit with X_L and X_C ,
 - A. I_L and I_C are 90° out of phase.
 - B. I_L and I_C are in phase.
 - C. I_L and I_C are 180° out of phase.
 - D. X_L and X_C are 90° out of phase.
2. What is the term for the out-of-phase power associated with inductors and capacitors?
 - A. Effective power.
 - B. True power.
 - C. Peak envelope power.
 - D. Reactive power.
3. In an AC circuit, if the voltage and current are in phase, the circuit is
 - A. resistive
 - B. capacitive
 - C. reactive
 - D. inductive
4. For a capacitor in a sine-wave ac circuit,
 - A. V_C lags i_C by 90°
 - B. i_C leads V_C by 90°
 - C. i_C and V_C have the same frequency.
 - D. All of the above.
5. Calculate the r.m.s. value of a sinusoidal current of maximum value 20A.
 - A. 7.07 A
 - B. 14.14 A
 - C. 28.28 A
 - D. 26.25 A
6. An alternating voltage is given by $v = 75 \sin(200\pi t - 0.25)$ volts. Find the phase angle (in degrees and minutes)
 - A. 14.32 (lagging)
 - B. 14.32 (leading)
 - C. 31.83 (lagging)
 - D. 31.83 (leading)
7. In a parallel RL circuit,
 - A. I_L lags I_R by 90° .
 - B. I_L leads I_R by 90° .
 - C. I_L and I_R are in phase.
 - D. I_R lags I_L by 90° .
8. Calculate the current taken by a $10 \mu\text{F}$ capacitor when connected to a 200 V, 100 Hz supply.
 - A. 0.1257 A
 - B. 125.7 A
 - C. 12.57 A
 - D. 1.257 A
9. What is the power factor (PF) of a purely resistive ac circuit?
 - A. 0.
 - B. 1.
 - C. 0.707.
 - D. Without values, it cannot be determined.
10. A resistive and a capacitive load of equal magnitude is connected in series, determine the phase difference between the voltage and the current.
 - A. current leads the voltage by 45°
 - B. current lags the voltage by 45°
 - C. current leads the voltage by 90°
 - D. current lags the voltage by 90°
11. For an inductor in a sine-wave ac circuit,
 - A. V_L leads i_L by 90° .
 - B. V_L lags i_L by 90° .
 - C. V_L and i_L are in phase.
 - D. None of the above.
12. What will happen when the power factor of a circuit is increased?
 - A. Reactive power increases
 - B. Active power increases
 - C. Both active and reactive powers increases
 - D. Both active and reactive powers decreases
13. Calculate the capacitive reactance of a capacitor of $20 \mu\text{F}$ when connected to an a.c. circuit of frequency 20 Hz
 - A. 397.9 ohms
 - B. 379.8 ohms
 - C. 399.7 ohms
 - D. 378.9 ohms
14. The impedance in the study of electronics is represented by resistance and _____.
 - A. inductance and capacitance
 - B. inductance
 - C. reactance
 - D. capacitance
15. In a series RLC circuit,
 - A. X_L and X_C are 180° out of phase.
 - B. I_L and I_C are 180° out of phase.
 - C. I_L and X_C are 180° out of phase.
 - D. I_L and X_C are in phase.
16. A resistive and a capacitive load of equal magnitude is connected in series, determine the phase difference between the voltage and the current.
 - A. current leads the voltage by 45°
 - B. current lags the voltage by 45°
 - C. current leads the voltage by 90°
 - D. current lags the voltage by 90°
17. In a series RC circuit,
 - A. V_C leads V_R by 90°
 - B. V_C and I are in phase.
 - C. V_C lags V_R by 90°
 - D. Both B and C.
18. An e.m.f. of 200 V at a frequency of 2 kHz is applied to a coil of pure inductance 50 mH. Determine the reactance of the coil and the current flowing in the coil.
 - A. 628Ω ; 0.318 A
 - B. 628Ω ; 0.328 A
 - C. 630Ω ; 0.318 A
 - D. 618Ω ; 0.318 A
19. If a circuit has an admittance of $Y = 0.2 + j0.6$, the circuit is
 - A. purely inductive
 - B. inductive
 - C. capacitive
 - D. reactive
20. What is the significance of connecting loads in parallel?
 - A. It makes power consumption less
 - B. It provides greater efficiency
 - C. It increases the safety factor



- D. It allows independent operations of loads
21. What is the term used to identify an AC voltage that would cause the same heating in a resistor as a corresponding value of DC voltage?
- A. Cosine voltage.
B. Power factor.
C. Root mean square (RMS).
D. Average voltage.
22. When the frequency of the applied voltage increases in a series RC circuit,
- A. the phase angle, θ , becomes more negative.
B. Z_T increases.
C. Z_T decreases.
D. Both A and C.
23. The voltages across three components in a series circuit when connected across an a.c. supply are:
- $$v_1 = 30 \sin \left(300\pi t - \frac{\pi}{6} \right) \text{ volts,}$$
- $$v_2 = 40 \sin \left(300\pi t + \frac{\pi}{4} \right) \text{ volts and}$$
- $$v_3 = 50 \sin \left(300\pi t + \frac{\pi}{3} \right) \text{ volts.}$$
- Calculate the supply voltage in sinusoidal form.
- A. $V_s = 97.39 \sin(300\pi t + 0.620) \text{ V}$
B. $V_s = 97.93 \sin(300\pi t + 0.620) \text{ V}$
C. $V_s = 79.39 \sin(300\pi t + 0.620) \text{ V}$
D. $V_s = 97.39 \sin(300\pi t + 0.602) \text{ V}$
24. The product of the readings of an AC voltmeter and AC ammeter is called:
- A. Apparent power.
B. True power.
C. Power factor.
D. Current power.
25. A resistor of 150Ω , a coil with reactance 100Ω and a capacitor with reactance -200Ω are connected in series. What is the complex impedance $R + jX$?
- A. $150 + j100$
B. $150 - j200$
C. $100 - j200$
D. $150 - j100$
26. In an ac circuit with only series capacitors,
- A. V_T leads I_T by 90° .
B. V_T lags I_T by 90° .
C. each capacitor voltage drop leads I_T by 90° .
D. Both A and C.
27. A value of $-j500 \Omega$ represents
- A. 500Ω of inductive reactance.
B. 500Ω of capacitive reactance.
C. 500Ω of resistance.
D. 500Ω of conductance.
28. Inductive reactance, X_L ,
- A. applies only to non-sinusoidal waveforms or dc.
B. applies only to sine waves.
C. applies to either sinusoidal or non-sinusoidal waveforms.
D. is inversely proportional to frequency.
29. Susceptance, B , is the
- A. reciprocal of impedance.
B. reciprocal of reactance.
C. reciprocal of resistance.
D. same as conductance.
30. What is the total impedance of a series AC circuit having a resistance of 6 ohms, an inductive reactance of 17 ohms, and zero capacitive reactance?
- A. 6.6Ω
B. 18Ω
C. 11Ω
D. 23Ω
31. When the frequency of the applied voltage decreases in a parallel RL circuit,
- A. the phase angle, θ_i , becomes less negative.
B. Z_{EQ} increases.
C. Z_{EQ} decreases.
D. Both A and B.
32. An impedance draws a current $i = 10 \cos(\omega t - 30^\circ)$ from a voltage, $v = 220 \sin(\omega t + 30^\circ)$. What is the impedance?
- A. $15.6 - j15.6$
B. $15.6 + j15.6$
C. $19.1 - j11.1$
D. $11.1 + j19.1$
33. In a parallel RC circuit,
- A. I_C lags I_R by 90°
B. I_R and I_C are in phase.
C. I_C leads I_R by 90° .
D. I_R leads I_C by 90° .
34. If a circuit has an admittance of $Y = 0.5 - j0.9$, the circuit is
- A. resistive
B. capacitive
C. inductive
D. reactive
35. A coil, resistor and capacitor are in parallel. The resistance is 1Ω ; the capacitive susceptance is 1.0 siemens; the inductive susceptance is -1.0 siemens. Then the frequency is cut to half its former value. What will be the admittance vector, $G + jB$, at the new frequency?
- A. $1 + j0$
B. $1 - j1.5$
C. $1 + j1.5$
D. $1 - j2$
36. A branch current of $+j250 \text{ mA}$ represents
- A. 250 mA of inductive current.
B. 250 mA of resistive current.
C. 250 mA of capacitive current.
D. 250 mA of in-phase current.
37. A capacitor has a capacitive reactance of 400Ω when connected to a 100 V , 25 Hz supply. Determine its capacitance and the current taken from the supply.
- A. $15.92 \mu\text{F}$; 0.25 A
B. $15.92 \mu\text{F}$; 0.15 A
C. $15.29 \mu\text{F}$; 0.25 A
D. $15.92 \mu\text{F}$; 0.025 A
38. What happens to reactive power in a circuit that has both inductors and capacitors?



- A. It is dissipated as heat in the circuit.
B. It alternates between magnetic and electric fields and is not dissipated.
C. It is dissipated as inductive and capacitive fields.
D. It is dissipated as kinetic energy within the circuit.
39. How do you compute true power (power dissipated in the circuit) in a circuit where AC voltage and current are out of phase?
- A. Multiply RMS voltage times RMS current.
B. Subtract apparent power from the power factor.
C. Divide apparent power by the power factor.
D. Multiply apparent power times the power factor.
40. In a parallel RC circuit,
- A. V_A and I_R are in phase.
B. V_A and I_C are in phase.
C. I_C and I_R are in phase.
D. V_A and I_R are 90° out of phase.
41. An alternating voltage given by $v = 100 \sin 240t$ volts is applied across a coil of resistance 32Ω and inductance 100 mH . Determine the circuit impedance.
- A. 40 ohms C. 20 ohms
B. 50 ohms D. 30 ohms
42. What is a sine wave?
- A. A constant-voltage, varying-current wave.
B. A wave whose amplitude at any given instant can be represented by the projection of a point on a wheel rotating at a uniform speed.
C. A wave following the laws of the trigonometric tangent function.
D. A wave whose polarity changes in a random manner.
43. A 120 mH inductor has a 50 mA , 1 kHz alternating current flowing through it. Find the p.d. across the inductor
- A. 36.7 V C. 35.7 V
B. 33.7 V D. 37.7 V
44. What type of wave is characterized by a rise time significantly faster than the fall time (or vice versa)?
- A. Cosine wave.
B. Square wave.
C. Sawtooth wave.
D. Sine wave.
45. What type of wave is made up of sine waves of the fundamental frequency and all the odd harmonics?
- A. Square. C. Sine.
B. Cosine. D. Tangent.
46. When the frequency of the applied voltage increases in a parallel RC circuit,
- A. the phase angle, θ , increases.
B. Z_{EQ} increases.
C. Z_{EQ} decreases.
D. Both A and C.
47. How many degrees are there in one complete sine wave cycle?
- A. 90 degrees.
B. 270 degrees.
C. 180 degrees.
D. 360 degrees.
48. In a series RC circuit,
- A. V_C and V_R are in phase.
B. V_T and I are always in phase.
C. V_R and I are in phase.
D. V_R leads I by 90° .
49. What is the description of a square wave?
- A. A wave with only 300 degrees in one cycle.
B. A wave whose periodic function is always negative.
C. A wave whose periodic function is always positive.
D. A wave that abruptly changes back and forth that abruptly changes back and forth between two voltage levels and stays at these levels for equal amounts of time.
50. What type of wave is made up of sine waves at the fundamental frequency and all the harmonics?
- A. Sawtooth wave.
B. Square wave.
C. Sine wave.
D. Cosine wave.
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